For each of the following, let $s(t) = 1 + \dfrac{2}{\sqrt{t+1}}$ represent the position function of some object. Compute the average velocity over each given time interval.

1. Time Interval: $[0, 3]$
2. Time Interval: $[8, 15]$
3. Time Interval: $[5, 8]$
4. Time Interval: $[0, 2]$
5. Time Interval: $[0, h]$
6. Time Interval: $[3, 3 + h]$

For all of the following problems, assume that there is object, $\mathcal{O}$, where its position is represented by $s(t)$ (meters) after t (seconds) have passed and its velocity is represented by $v(t)$ (meters per second) after t (seconds) have passed.

7. Precisely, translate the mathematical expression $s(2) = 5$ into english (include units in your response).
8. Precisely, translate the mathematical expression $s(0) = 10$ into english (include units in your response).
9. Precisely, translate the mathematical expression $v(1.1) = 5.3$ into english (include units in your response).
10. Give the mathematical expression for $\mathcal{O}$'s position after 3s have passed.
11. Give the expression that represents $\mathcal{O}$'s average velocity between 3 and 7s.
12. Give the expression that represents $\mathcal{O}$'s velocity after 7s.
13. Suppose $\mathcal{O}$'s initial position was a (meters). Give the expression that represents $\mathcal{O}$'s initial position.
14. Suppose $\mathcal{O}$'s initial position was a (meters) and 5s later was 17m. Find $\mathcal{O}$'s average velocity during the first five seconds.

Answer the following.

15. Let $s(t) = -16t^2 + 100t$ be the position function (in meters) of some object after t seconds
    (a) Complete the following table of average velocities.

Time Interval	**Average Velocity**
$[3, 3.1]$	
$[3, 3.01]$	
$[3, 3.001]$	
$[3, 3.0001]$	

    (b) Give a real world interpretation of the value in the last row of the table (within the context of this section)?

16. Let $s(t) = e^{-t^2}$ be the position function (in meters) of some object after t seconds.
    (a) Complete the following table of average velocities.

Time Interval	Average Velocity
$[1, 1.1]$	
$[1, 1.01]$	
$[1, 1.001]$	
$[1, 1.0001]$	

(b) Give a real world interpretation of the value in the last row of the table (within the context of this section)?

17. Given the position function, $s(t) = \sin\left(\frac{t}{3}\right)$ (meters), estimate the instantaneous velocity at $t = 1$s to within 3 decimal places.

18. Given the position function, $s(t) = \frac{10}{1+t}$ (meters), estimate the slope of the tangent line at $t = -4$s to within 2 decimal places.

Answer the following conceptual questions.

19. Let $f(x)$ be a position function of some object. If P a point on the graph of $f(x)$, interpret the meaning of P.

20. Let $f(x)$ be a position function of some object and P a point on the graph of $f(x)$. Explain the relationship between velocity of the object at P and the tangent line of $f(x)$ through P.

21. In general, explain the relationship between velocity and tangent lines.

22. Let $f(x)$ be a position function of some object and P a point on the graph of $f(x)$. Explain the relationship between rate of change of the object at P and velocity of the object at P.

23. In general, explain one relationship between secant lines and tangent lines on a function $f(x)$ (in the context of this section).

24. Suppose an object's position function is given by $s(t)$. Explain how average velocity could be used to estimate the velocity of an object after 6s.

For each of the following, let $s(t) = 1 + \dfrac{2}{\sqrt{t+1}}$ represent the position function of some object. Compute the average velocity over each given time interval.

1. $\bar{v}_{0\to 3} = \frac{s(3)-s(0)}{3-0} = -\frac{1}{3}$

3. $\bar{v}_{5\to 8} = \frac{s(8)-s(5)}{4} = \frac{6-2\sqrt{6}}{\sqrt{6}}$

5. $\bar{v}_{0\to h} = \frac{s(h)-s(0)}{h-0} = \frac{-2\sqrt{h+1}+2}{h\sqrt{h+1}}$

For all of the following problems, assume that there is object, $\mathcal{O}$, where its position is represented by $s(t)$ (meters) after t (seconds) have passed and its velocity is represented by $v(t)$ (meters per second) after t (seconds) have passed.

7. After 2 seconds, the position of $\mathcal{O}$ is 5 meters

9. After 1.1 seconds, the velocity of $\mathcal{O}$ is 5.3 meters per second

11. $\frac{s(7)-s(3)}{7-3} = \frac{s(7)-s(3)}{4}$

13. $s(0) = a$

Answer the following.

15. (a)

Time Interval	Average Velocity
$[3, 3.1]$	2.4
$[3, 3.01]$	3.84
$[3, 3.001]$	3.984
$[3, 3.0001]$	3.9984

;

(b) The velocity of the object after 3 seconds is approximately 3.9984.

17. 0.314 m/s

Answer the following conceptual questions.

19. P corresponds to a specific position at some time where the y-coordinate of P is the position and x-coordinate of P is the time it attains that position.

21. When the function in question is a position function of an object, the slope of any tangent line the function through a point (say (x_1, y_1)) can be interpreted as the velocity of the object at the time $= x_1$

23. Secant lines can be used to estimate tangent lines. For example, to find a tangent line through the point $(6, f(6))$, one could instead find the secant line through points $(6, f(6))$ and $(6.0001, f(6.0001))$. The same technique is used to estimate the slope of a tangent line.